

Question 1. Who is the president of the USA?

A: Donald Trump	B: Joe Biden	C: Bill Clinton	D: George Bush	E: Barack Obama
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Question 2. What is $2 + 2$?

A: 3	B: 1	C: 2	D: 4	E: 5
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Question 3. Which sorting algorithm typically has $O(n^2)$ average time complexity?

A: Bubble Sort	B: Merge Sort	C: Quick Sort	D: Insertion Sort	E: Radix Sort
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Question 4. What is the derivative of x^2 ?

A: x^3	B: 1	C: $2x$	D: $\frac{1}{x}$	E: $4x^2$
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Question 5. Compute $\int e^{2x} dx$.

A: $\frac{1}{2}e^{2x}$	B: e^{2x}	C: xe^{2x}	D: $2e^{2x}$	E: e^x
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Question 6. Evaluate $\lim_{x \rightarrow 0} \frac{\sin x}{x}$.

A: 1	B: 0	C: 2	D: ∞	E: Does not exist
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Question 7. Which is the correct factorization of $x^3 - y^3$?

A: $(x - y)(x^2 + xy + y^2)$	B: $(x + y)(x^2 - xy + y^2)$
C: $(x^3 - y^3)(x - y)$	D: $(x - y)(x + y)$
E: $(x - y)(x^2 + xy + y^2)$	

Question 8. Evaluate the following statements regarding advanced calculus topics.

A: An exceptionally long option that exceeds half of the text width to test text wrapping within the designated rectangle.
B: Another lengthy option that requires the rectangle to be enlarged to accommodate the extended text without overflow.
C: A standard short option.
D: A moderately long option to ensure that the rectangle adjusts its size appropriately.
E: Yet another long option that tests the flexibility of the rectangle sizing in the LaTeX document.

Question 9. Which function is its own derivative?

A: $f(x) = x^2$	B: $f(x) = \sin x$	C: $f(x) = e^x$	D: $f(x) = \ln x$	E: $f(x) = \sqrt{x}$
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Question 10. Which of the following statements are true about calculus functions?

A: The integral of $\frac{1}{x}$ is $\ln x + C$.	B: The derivative of e^x is e^x .
C: The limit of $\frac{\sin x}{x}$ as $x \rightarrow 0$ is 1.	D: The derivative of $\ln x$ is $\frac{1}{x}$.
E: The integral of x is $\frac{1}{2}x^2 + C$.	

Question 11. Answer the following advanced calculus problems.

A: Given the function $f(x) = x^3 - 3x + 2$, find all real roots and verify them by substitution.
B: Determine the convergence of the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ using the p-test.
C: Compute the limit $\lim_{x \rightarrow \infty} \frac{e^x}{x^n}$ for any positive integer n .
D: Find the second derivative of $f(x) = \sin(x)$ and interpret its geometric meaning.
E: Evaluate the definite integral $\int_0^{\pi} \sin(x) dx$ and explain its significance.